

Alejandro Parga MD PhD

Physician-Scientist | Biomedical Innovation and Program Strategy Leader

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Profile

Physician-scientist and translational biomedical leader with 15+ years of experience spanning clinical medicine, neurotechnology, and therapeutic innovation across academic, federal, military, and healthcare environments. Experienced in architecting multidisciplinary research programs, building technical infrastructure, coordinating cross-institutional teams, and translating emerging technologies into clinically relevant biomedical platforms.

Leadership experience managing \$10M+ in research and infrastructure allocations, coordinating multidisciplinary teams of 6–12 investigators, and contributing to 7 NIH-, DoD-, and VA-funded programs. Track record includes laboratory and core-facility development, scientific platform implementation, federal research collaboration, and technology convergence across neurophysiology, multiomics, imaging, circuit engineering, and photomodulation.

Particularly focused on high-risk, high-reward biomedical opportunities where convergence of emerging technologies can create new therapeutic paradigms and accelerate measurable improvements in human health.

Scientific Architect: combines scientific depth, operational leadership, and clinical perspective to build teams, develop programs, deploy resources, and translate unconventional biomedical concepts into executable strategies with potential for transformative impact.

Leadership at a Glance

15+ years biomedical R&D • \$10M+ research/funding portfolio managed • \$4.05M Infrastructure investments • 7 NIH/DoD/VA -funded initiatives • 12+ investigators coordinated • 16+ medical professionals supervised • 1500-person military medical population served • 4+ research platforms developed

Leadership and Program Management

- **Program Leadership:** Lead multidisciplinary biomedical research programs spanning neuroscience, neurophysiology, molecular biology, multiomics, imaging, neurotechnology, and translational therapeutic development, coordinating teams of **6–12 investigators** from postdoctoral scientists and laboratory personnel to principal investigators and physicians.
- **Team & Institutional Leadership:** Coordinate multidisciplinary teams across academic, VA, DoD, military, and healthcare research environments. As Medical Coordinator, supervised **specialized surgical teams and 16+ medical professionals** while coordinating clinical operations for a military medical center serving approximately **1,500 personnel**.
- **Federal Program Experience:** NIH/NINDS • NIMH • VA • DoD collaborations. Contributed to **7 federally funded programs across NIH-, NINDS-, NIMH-, and VA**, serving as research lead, electrophysiology lead, co-investigator, or technical specialist across multidisciplinary projects.

- **Capital & Infrastructure Management:** Contributed to approximately **\$4.05M in scientific infrastructure and equipment investments**, including **\$100K** for postdoctoral laboratory expansion, **\$250K** VA-UW electrophysiology infrastructure, **\$1.2M** for advanced microscopy and imaging infrastructure, and **\$1.5M** for Seattle Children's laboratory equipment, remodeling, and implementation of a new neurodevelopmental research laboratory.
- **Program & Platform Development:** Established and expanded research infrastructure supporting electrophysiology, advanced microscopy, circuit mapping, and neurodevelopmental research. Developed and coordinated research portfolios spanning **neuroanatomical vector exploration, non-human primate single-cell neurocharacterization, and photodynamic platforms.**
- **Scientific Strategy & Execution:** Translate emerging technologies and mechanistic discoveries into executable research programs, technical platforms, milestones, and translational objectives, aligning scientific opportunities with available infrastructure, expertise, and resources.
- **Technology Convergence:** Integrate electrophysiology, Patch-seq, circuit tracing, transcriptomics, advanced imaging, optogenetics, photomodulation, and computational approaches to develop new strategies for understanding and therapeutically modulating brain function.
- **Translation & Innovation:** Connect fundamental neuroscience with clinical needs and therapeutic development, with emphasis on traumatic brain injury, neural circuit dysfunction, neurodevelopment, precision neuromodulation, and emerging therapeutic technologies.

Selected Programs & Platforms

Neuroanatomical Vector Exploration

Development and evaluation of vector-based approaches for neuronal circuit mapping and targeted manipulation.

Non-human primate Single-Cell Neurocharacterization

Integration of NHP electrophysiology, Patch-seq, and multiomics to characterized neuronal populations.

Photomodulation Platform

Development of spatially and temporally controlled approaches for neural modulation and therapeutic intervention

Traumatic Brain Injury & Neural Repair

Multimodal investigation of circuit dysfunction, inhibitory signaling, and therapeutic modulation following brain injury.

Professional Experience

Senior Scientific Lead

Seattle Children's Research Institute — Seattle, WA
2023–2026

- Established and orchestrated a new neurodevelopmental research laboratory, including space remodeling, scientific infrastructure, and acquisition of \$1.5M in research equipment.
- Built multidisciplinary collaborations spanning Seattle Children's, University of Washington, Fred Hutch, and federal research partners.
- Developed research programs integrating neurophysiology, molecular profiling, imaging, circuit mapping, and photomodulation.

- Hired and coordinated 12 investigators and technical personnel across complementary scientific cores.
- Developed translational research strategies connecting mechanistic neurodevelopmental biology with therapeutic targets and patient relevant biomarkers.
- Identified and evaluated opportunities for technology convergence across neurophysiology, multiomics, circuit mapping, and precision neuromodulation.

Scientist & Senior Fellow

VA Puget Sound Health Care System & University of Washington — Seattle, WA
2017–2023

- Operationalized and electrophysiology research core within the VA Puget Sound Health Care System, including \$250K in equipment configuration and implementation of optical stimulation and circuit-mapping capabilities.
- Expanded Collaborations across VA and University of Washington laboratories, enabling multidisciplinary research spanning TBI, epilepsy, GABAergic signaling, circuit physiology, and neurotechnology.
- Directed neurophysiology and multiomic research initiatives within the Epilepsy Center of Excellence, coordinating scientific activities across NIH- and DoD-supported laboratories.
- Developed experimental strategies integrating electrophysiology, circuit tracing, transcriptomics, imaging, and computational analysis to characterize disease-associated changes in neuronal networks.
- Investigated mechanisms of traumatic brain injury-induced network dysfunction and altered GABAergic signaling, including work published in *eNeuro*.
- Advanced Patch-seq and circuit-mapping platforms to connect neuronal molecular identity with morphology and functional excitability.
- Coordinated multidisciplinary research activities spanning experimental neuroscience, clinical research, neurotechnology, and computational biology.
- Contributed to strategic development of translational research programs linking fundamental mechanisms to therapeutic opportunities.
- Mentored trainees and interdisciplinary collaborators in neurophysiological experimentation, quantitative analysis, and translational neuroscience.

Postdoctoral Research Associate

University of Arizona, College of Medicine — Phoenix, AZ
2015–2016

- Developed photodynamic and targeted drug-delivery strategies for therapeutic intervention following traumatic brain injury.
- Integrated computational modeling with in vivo electrophysiology to evaluate mechanisms of neural dysfunction and recovery.
- Explored technology-enabled approaches for spatially and temporally controlled therapeutic modulation of neural systems.
- Collaborated across experimental and computational disciplines to develop translational neurotrauma research strategies.

Research Associate

Barrow Neurological Institute / Arizona State University — Phoenix, AZ
2008–2014

- Conducted multidisciplinary neuroscience research investigating neuronal circuits underlying sensory processing, motor function, and neuropsychiatric disease.

- Investigated corticostriatal, dopaminergic, and thalamocortical circuits using viral tracing, electrophysiology, stereotaxic techniques, and pharmacological approaches.
- Developed experimental models integrating circuit mapping and functional neurophysiology to interrogate distributed neuronal networks.
- Conducted pharmacological and systems-neuroscience studies leading to peer-reviewed publications, including work in *Biological Psychiatry*.
- Collaborated with investigators across neuroscience, medicine, and neuroengineering disciplines.

Medical Officer & Clinical Operations Coordinator Physician

La Arandia Military Base, US-supported Colombian Army — Florencia, Colombia

2006–2007

- Coordinated medical operations for a military population of approximately **1,500 personnel**, including emergency care, evacuation, trauma management, and surgical support.
- Supervised **3 physicians and 16 medical professionals** and coordinated multidisciplinary clinical operations.
- Managed pharmacy procurement and resource allocation based on epidemiological requirements.
- Directed emergency response and medical support under operationally constrained conditions.

Education

PhD, Neuroscience

Barrow Neurological Institute / Arizona State University — Phoenix, AZ

2014

Dissertation: *Cortical auditory functional activation by cortico-striato-thalamo-cortical circuits*

MD, Neurophysiology Focus

National University of Colombia — Bogotá, Colombia

2005

Trauma & Emergency Medicine Training

Larandia Military Base — Florencia, Colombia

2007

Industrial University of Santander — Bucaramanga, Colombia

2006

Clinical clerkships: San Rafael Hospital; Ramón González Valencia University Hospital

Selected Publications

- **Areal specializations in the morpho-electric and transcriptomic properties of layer 5 extratelencephalic-projecting neurons in the primate neocortex.** *Neuron*, 2024.
- **Traumatic brain injury broadly affects GABAergic signaling in dentate gyrus granule cells.** *eNeuro*, 2021.
- **Excessive striatal dopamine activates the auditory cortex via striato-thalamo-cortical projections.** *Biological Psychiatry*, 2016.

Technical Domain Expertise

Neurotechnology: Electrophysiology • Patch-seq • opto/chemogenetics • circuit mapping • photomodulation

Molecular & Cellular Technologies: Single-cell/single-nucleus transcriptomics • viral vectors • molecular profiling

Imaging & Morphological Analysis: Confocal • multiphoton • neuronal morphometry

Circuit Engineering & Experimental Platforms: Targeted neuromodulation • pharmacological intervention • photodynamic/photocaged therapeutic strategies

Computational & Quantitative Analysis: Python • R • Git • quantitative electrophysiology • transcriptomics • multidimensional data integration